

Engineering Mathematics: Total Derivatives & Modified Euler's Theorems

When working with multi-variable calculus, variables often depend on each other through intermediate paths, or functions are wrapped inside non-linear operations (like $\log$, $\sin^{-1}$, or $\tan^{-1}$). This set of notes covers **Total Derivatives (Chain Rule with Tree Diagrams)** and the **Modified (Composite) Euler's Theorems** used to solve these systems efficiently.

1. Total Derivatives & Composite Functions (Chain Rule)

When a variable z depends on x and y , and those variables themselves depend on other parameters (like u and v), we use the multi-variable chain rule.

To prevent tracking errors, we map the dependencies using a **Tree Diagram**.

Case A: $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$

Here, z ultimately depends on a single independent variable t . The derivative is called the **Total Derivative** ($\frac{dz}{dt}$):

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Case B: $z = f(x, y)$ where $x = x(u, v)$ and $y = y(u, v)$

Here, z ultimately depends on two independent variables, u and v . We must use partial derivatives ($\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$):

$$\begin{aligned}\frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \\ \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}\end{aligned}$$

Visualizing with a Tree Diagram

To write these equations automatically without memorization:

1. Start at the top variable (z).
 2. Draw branches downward to its immediate intermediate variables (x and y). Write partial derivatives ($\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$) along these branches.
 3. From x and y , draw branches down to the final independent variables (u and v). Write their respective derivatives along those paths.
 4. **Rule:** Multiply down a single branch path; add distinct branch paths together.
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2. First-Order Modified Euler's Theorem

The Scenario

You are given a function equation $u = F(x, y)$, but u is **not** homogeneous because it is trapped inside a non-linear wrapper (e.g., $u = \log(\dots)$ or $u = \tan^{-1}(\dots)$).

However, if you isolate the algebraic core by transforming it into a substitute function $z = f(u)$, you find that z **is** homogeneous with a degree of n .

The Formula

Instead of working out the tedious partial derivatives of the wrapper manually, you can compute the first-order differential expression directly using:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)} = g(u)$$

Where:

- $f(u)$ is the isolated algebraic function (z).
- $f'(u)$ is the derivative of that function with respect to u .
- n is the degree of homogeneity of z .
- $g(u)$ is the resulting simplified expression.

Common Shortcuts for $g(u)$

If the wrapper is...	Isolated form $f(u) = z$	Derivative $f'(u)$	Value of $g(u) = n \cdot \frac{f(u)}{f'(u)}$
$u = \log(\text{core})$	e^u	e^u	n
$u = \sin^{-1}(\text{core})$	$\sin u$	$\cos u$	$n \tan u$
$u = \cos^{-1}(\text{core})$	$\cos u$	$-\sin u$	$-n \cot u$
$u = \tan^{-1}(\text{core})$	$\tan u$	$\sec^2 u$	$n \sin u \cos u = \frac{n}{2} \sin 2u$

3. Second-Order Modified Euler's Theorem

The Formula

If a composite question requires you to find the value of a second-order differential expression, computing it directly by taking second derivatives of wrappers is incredibly error-prone.

Instead, find the first-order expression $g(u)$ from the step above, and plug it directly into this corollary:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) \cdot [g'(u) - 1]$$

Where:

- $g(u)$ is the calculation outcome from the First-Order Modified Theorem.
- $g'(u)$ is the derivative of that specific outcome expression with respect to u .

Step-by-Step Proof of the First-Order Modified Theorem

Let $z = f(u)$ be a homogeneous function of independent variables x and y with a degree of n .

By standard Euler's Theorem, we can state:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = nz \quad \text{— (Equation 1)}$$

Since z is a function of u , and u is a function of x and y , we can find the partial derivatives of z using the single-variable composite Chain Rule:

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = f'(u) \frac{\partial u}{\partial x} \\ \frac{\partial z}{\partial y} &= \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = f'(u) \frac{\partial u}{\partial y} \end{aligned}$$

Substitute these Chain Rule values back into Equation 1:

$$x \left(f'(u) \frac{\partial u}{\partial x} \right) + y \left(f'(u) \frac{\partial u}{\partial y} \right) = nz$$

Factor out the shared derivative term $f'(u)$ from the left side:

$$f'(u) \cdot \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = nz$$

Since our original setup defined $z = f(u)$, substitute $f(u)$ back in place of z :

$$f'(u) \cdot \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = n \cdot f(u)$$

Divide both sides by $f'(u)$ to isolate the primary operator expression:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)}$$

Hence, proved.